

Speech and Audio Processing

10 - speech system building

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Class schedule (tentative)

01	Oct. 03	Introduction	Learn from basic to state-of-the-art about speech processing.
02	Oct. 10	Speech production / 音声生成	
03	Oct. 17	Speech synthesis / 音声合成	
04	Oct. 24	Speech synthesis 2 / 音声合成2	
05	Oct. 31	Speech recognition / 音声認識	
06	Nov. 7	Self-supervised learning / 自己教師あり学習	
07	Nov. 14	Sound recognition/synthesis / 環境音認識合成	
08	Nov. 28	Foundation model / 基盤モデル	
09	Dec. 05	LLM and dialogue system / LLMと対話システム	
10	Dec. 12	Speech system building / 音システム構築	Learn how to build your speech system.
11	Dec. 19	Speech system building / 音システム構築	
12	Dec. 26	Speech system building / 音システム構築	
13	Jan. 09	Speech system building / 音システム構築	
14	Jan. 23	Final presentation / 最終発表	

Grades and themes

- Grades
 - 50 / 100 ... final presentation
 - 50 / 100 ... final report
- Perform a unique project that either
 - improves on state-of-the-art in speech-related tasks,
 - applies {machine learning, signal processing} to a unique task,
 - investigates human behaviors in human–speech-AI interaction.
 - prototypes the future of human and speech AI.
- Who to do with
 - Alone or team (5 students and the less per team)
 - Recommended to make a team with other lab's students

Final presentation on Jan. 23

- X-minute (as of now, $X=4$) oral presentation
- Talk about purpose, system, and evaluation of your project
- Language
 - Slides in English.
 - Talk in Japanese or English, w/ simultaneous translation.
- Slides
 - Edit shared Google slides. The link will be shared later.

Final report on Jan. 30 (TBA, later than presentation)

- 2-page PDF. (more pages are of course acceptable.)
 - Each team may submit the same report. (if you form a team)
 - Write about purpose, system, and evaluation of your project.
- Language
 - Japanese or English.
- No specific format
 - If you (or your team) plan to submit to domestic or international conferences, you are recommended to use the conference format.
 - Peer-reviewed: INTERSPEECH (1st tier), IEEE SLT (2nd tier)
 - Domestic: 音学シンポジウム (Otogaku, submission in Apr.)

Questions to me

- Either way is OK:
 - K-LMS
 - E-mail (shinnosuke_takamichi@keio.jp)
 - University-wide or takamichi-lab slack
 - Talk during this class time
 - I will be in this room every class.
- Ask me if you are struggling
 - to decide a project theme,
 - to find state-of-the-art methods,
 - to evaluate your system, or
 - to submit your project to a conference.

Project themes

Survey

- Conferences (to find existing tasks and methods)
 - Speech and audio: IEEE ICASSP, Interspeech, ASRU, SLT, etc.
 - Natural language: ACL, EMNLP, NAACL, etc.
 - Machine learning: NeurIPS, AAAI, ICML, ICLR, IJCAL, etc.
 - Computer vision: CVPR, ICCV, etc.
 - Human computer interaction: CHI, UIST, ICMI, etc.
- Demand (to find what we should solve by technologies)
 - What will be ordinary in our lives 10 years from now?
 - Who (or what) is waiting for your help?

Tips

- OK as theme
 - Speech or audio-related tasks.
 - Multi-modality but speech or audio must be included.
- Not OK as theme
 - Completely same to your master-course research.
 - Extending the research to a new speech/audio theme is OK.
- Subjective evaluation is very important.
 - When you treat generation tasks, subjective evaluation is a key.

1. improving on state-of-the-art (SoTA)

- Find papers (and **open-source codes**) of SoTA work.
- Find unsolved issues or trade-off of the work. For example,
 - it does not achieve human-quality, or
 - it achieves high-quality but heavy computation.
- Implement a suitable way.
 - Quality enhancing? Speeding up? light computation? Universality?
- Evaluate your system.
 - Objective evaluation should be performed.
 - Subjective evaluation is also important.

2. applying {machine learning, signal processing} to a unique task.

- Create a unique task.
 - **You should answer to “Why must we tackle the task?” “Who (or what) is benefited by solving the task?”**
 - Hints to find a unique task exist in your lives.
- Construct a new dataset (If there exists no data).
 - Copyrighted work (e.g., book) and web data (e.g., YouTube) are allowed to use, but do to redistribute the data to other people.
- Apply existing models and evaluation methods to the task.
 - Create new models or evaluation (if needed).
 - Analyze the unresolved points.

3. investigating human behaviors in human–speech-AI interaction.

- Design scenes of human–speech-AI interaction.
 - **What does speech-AI mean to you?**
 - What is the difference from human-human interaction?
- Implement speech AI.
 - Minimal implementation is good.
- Observes human behavior.
 - Speech? Vision? Motion? Biological signals? Smartphone is a suitable.
- Analyze the behavior.
 - Difference from human-human communication.

4. prototyping the future of human and speech AI.

- Prototype our lives in 2035.
 - Values, cultures, etc.
- Implement a minimal prototype.
 - Be careful not to overextend the project.
- Do experiments and find a way to redesign the future.

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You can work from anywhere
(not limited to this room).

You have to come to this room.
(on-site only)

Theme examples (my past research)

Fully automated training data selection for speech synthesis

1. Improving on SoTA

2. unique task

3. Human-AI interaction

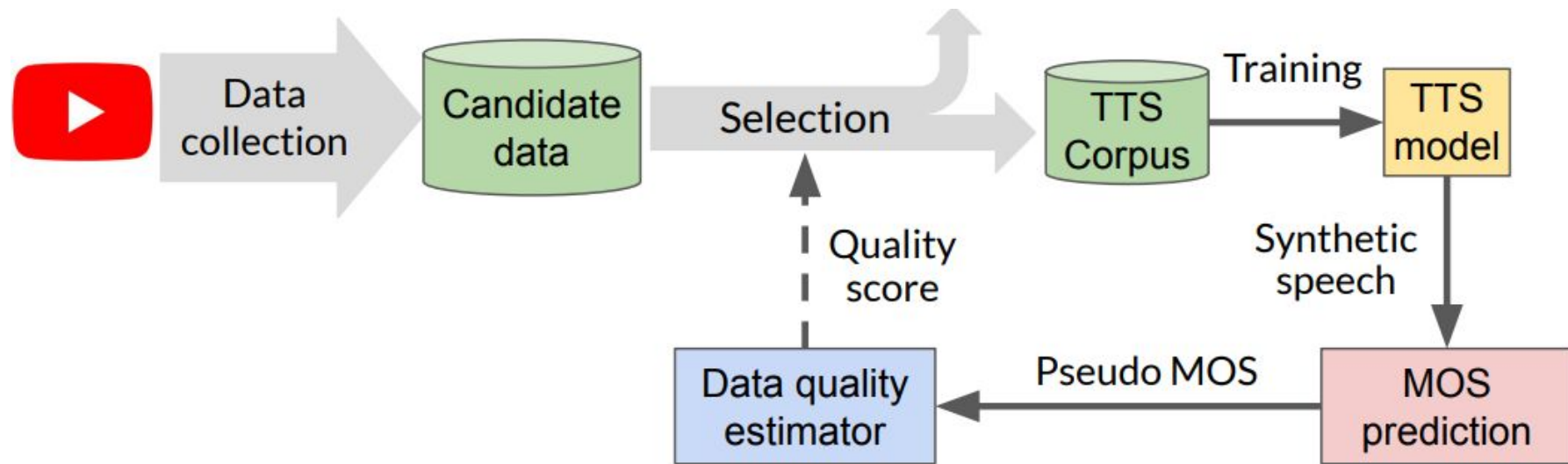
4. Prototyping future

TTSOps: A Closed-Loop Corpus Optimization Framework for Training Multi-Speaker TTS

Models from Dark Data

Kentaro Seki , [Shinnosuke Takamichi](#) , [Takaaki Saeki](#) , and [Hiroshi Saruwatari](#)

In IEEE/ACM Transactions on Audio, Speech, and Language Processing , Nov 2025



Analyzing alcohol-intoxicated singing voices

1. Improving on SoTA
2. **unique task**
3. Human-AI interaction
4. Prototyping future

高道 慎之介, 齋藤 大輔, "歌声合成に向けた酒酔い歌声の特徴量分析に関する初期検討," 日本音響学会2019年春季研究発表会講演論文集,

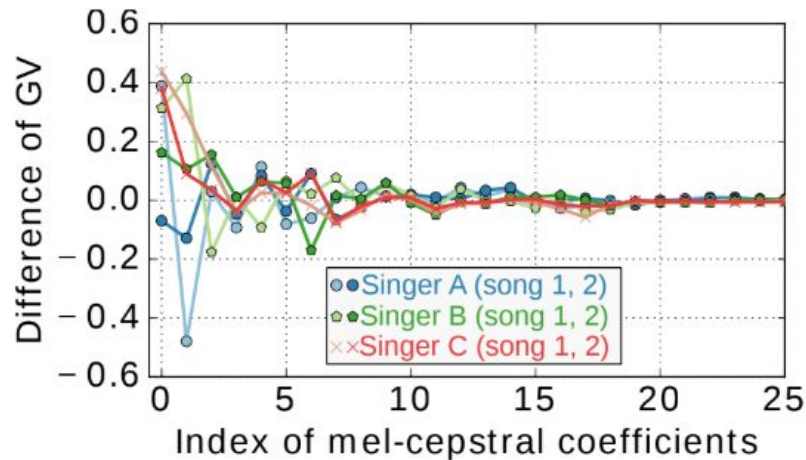


Fig. 1 飲酒前後におけるメルケプストラムのGVの差

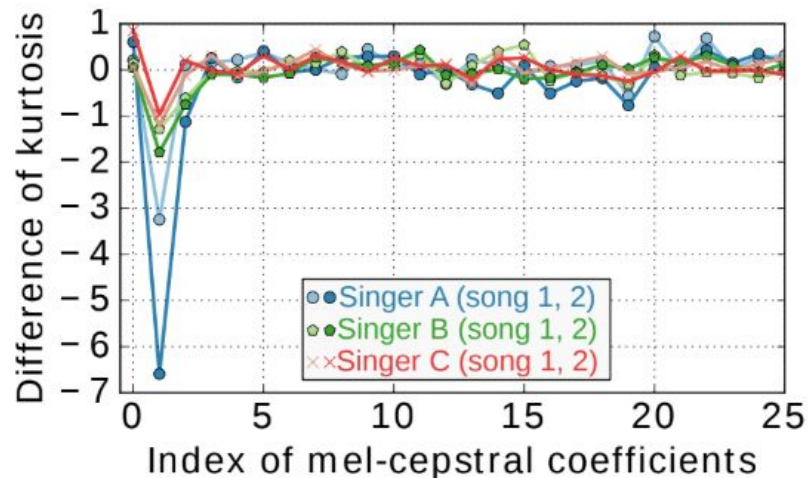


Fig. 2 飲酒前後におけるメルケプストラムのカートシスの差

Learning foreign languages with accented speech

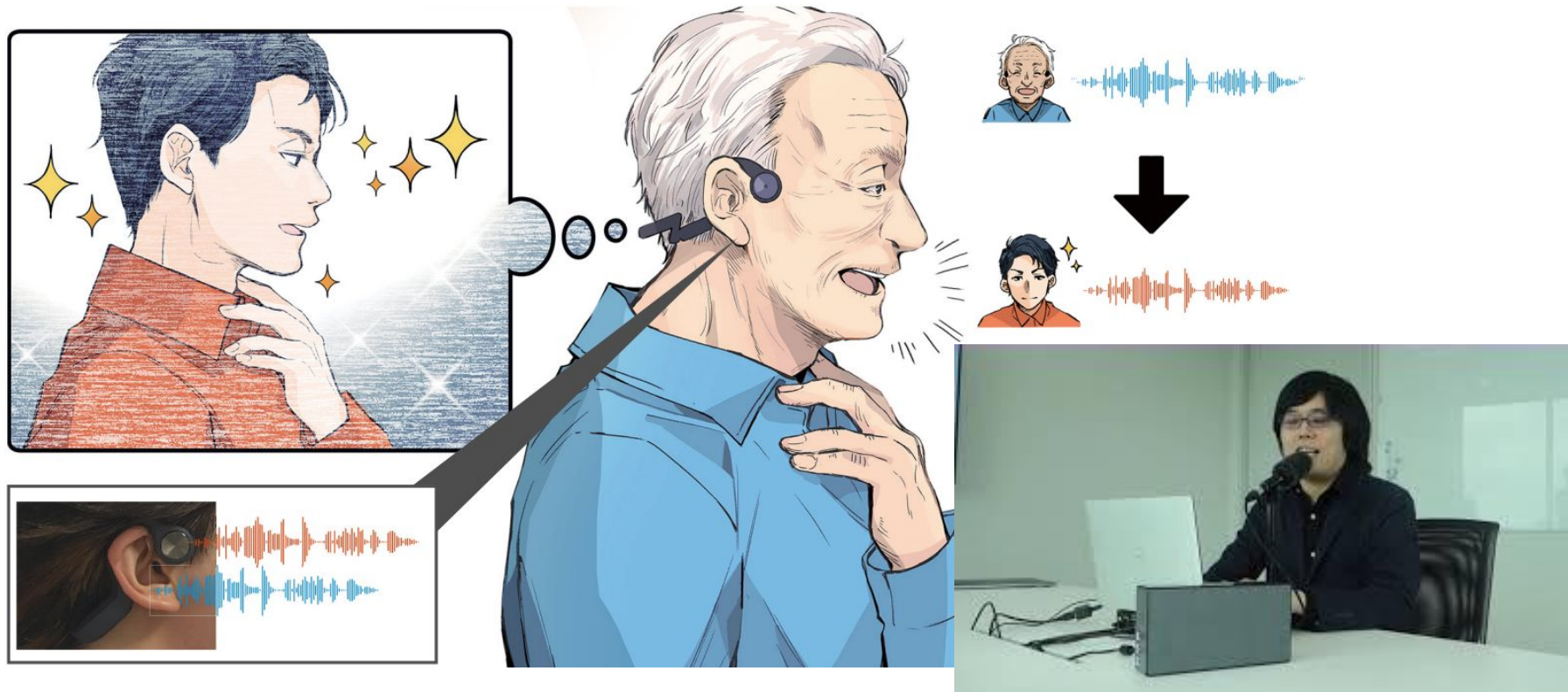
1. Improving on SoTA
2. unique task
3. **Human-AI interaction**
4. Prototyping future

3. Yasushige Ishikawa, Shinnosuke Takamichi, Yuta Matsunaga, Yusuke Yoshikawa, and Shigeo Fujiwara, "Speaking Practice Using Text-to-speech Technology: Japanese EFL Learners' Perceptions, " WorldCALL, Nov. 2023.



Digital speech makeup: transforming myself by changing self-voice

1. Improving on SoTA
2. unique task
3. Human-AI interaction
4. **Prototyping future**



Advice

- **Enjoying your project is all you need!**
 - Even if you cannot outperform SoTA, you won't fail this class.
 - Due to the time limitation, the number of participants in subjective evaluations may be small.
- You will earn credit of this class when
 - you tried to solve an issue in a technically correct way, or
 - you tried to define a valuable task.